
Re: Summer research update

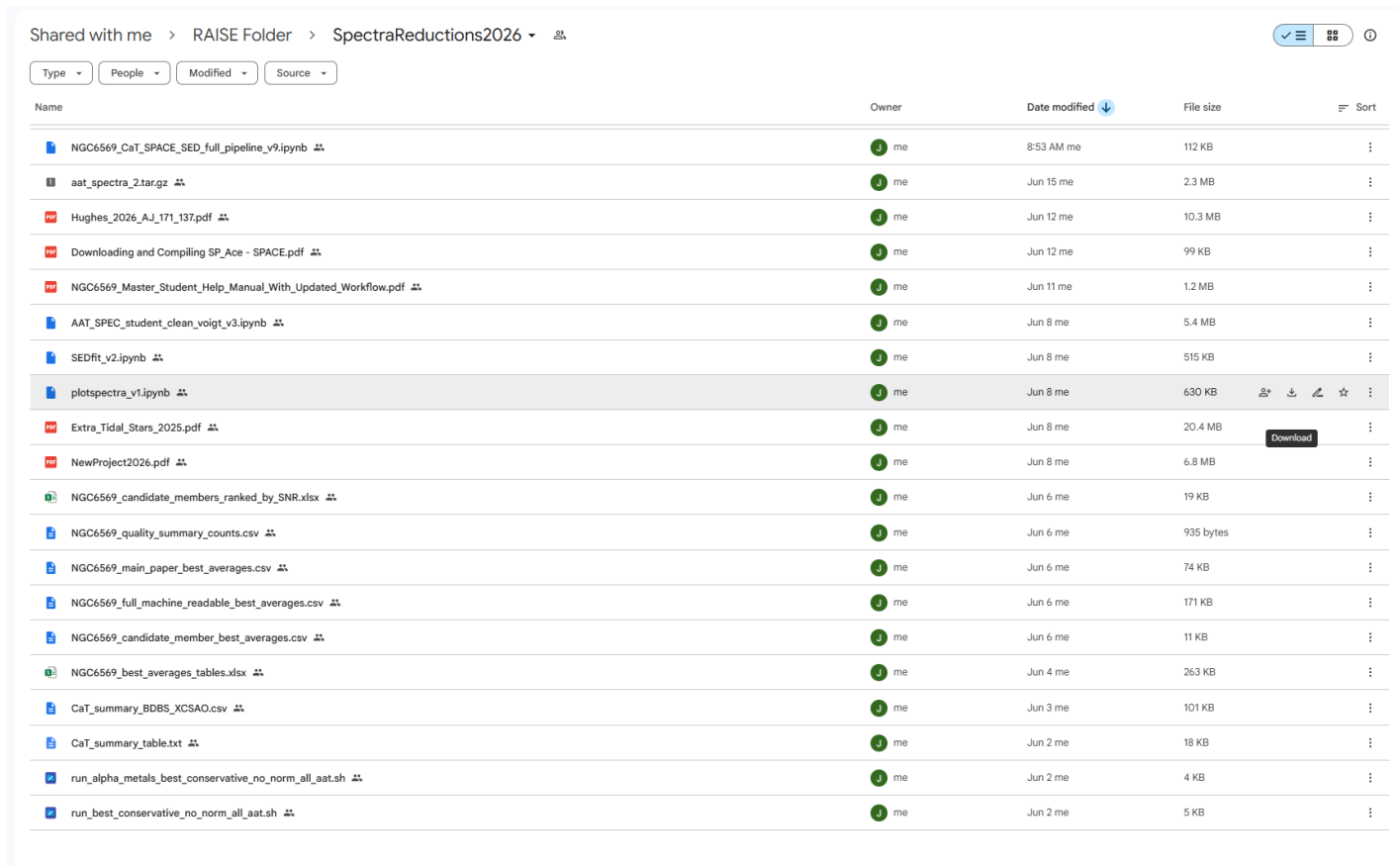
From Joanne Hughes Clark <jhughes@seattleu.edu>

Date Tue 6/16/2026 10:07 AM

To Joseph McEwen <jmcewen@seattleu.edu>; Payton Killcoyne <pkillcoyne@seattleu.edu>; Thea Campbell <tcampbell@seattleu.edu>; Laith Rabaa <lrabaa1@seattleu.edu>; Michael Williams <mwilliams8@seattleu.edu>; Keleb Mwaala <kmwaala@seattleu.edu>

Cc Kerry Cheon <kcheon@seattleu.edu>

The Google Drive directory should look like this. I've included all the training dataset. Make sure you can see all the files.



Shared with me > RAISE Folder > SpectraReductions2026

Type People Modified Source

Name	Owner	Date modified	File size	Sort
NGC6569_CaT_SPACE_SED_full_pipeline_v9.ipynb	me	8:53 AM me	112 KB	
aat_spectra_2.tar.gz	me	Jun 15 me	2.3 MB	
Hughes_2026_AJ_171_137.pdf	me	Jun 12 me	10.3 MB	
Downloading and Compiling SP_Ace - SPACE.pdf	me	Jun 12 me	99 KB	
NGC6569_Master_Student_Help_Manual_With_Updated_Workflow.pdf	me	Jun 11 me	1.2 MB	
AAT_SPEC_student_clean_voigt_v3.ipynb	me	Jun 8 me	5.4 MB	
SEDfit_v2.ipynb	me	Jun 8 me	515 KB	
plotspectra_v1.ipynb	me	Jun 8 me	630 KB	
Extra_Tidal_Stars_2025.pdf	me	Jun 8 me	20.4 MB	Download
NewProject2026.pdf	me	Jun 8 me	6.8 MB	
NGC6569_candidate_members_ranked_by_SNR.xlsx	me	Jun 6 me	19 KB	
NGC6569_quality_summary_counts.csv	me	Jun 6 me	935 bytes	
NGC6569_main_paper_best_averages.csv	me	Jun 6 me	74 KB	
NGC6569_full_machine_readable_best_averages.csv	me	Jun 6 me	171 KB	
NGC6569_candidate_member_best_averages.csv	me	Jun 6 me	11 KB	
NGC6569_best_averages_tables.xlsx	me	Jun 4 me	263 KB	
CaT_summary_BDBS_XCSAO.csv	me	Jun 3 me	101 KB	
CaT_summary_table.txt	me	Jun 2 me	18 KB	
run_alpha_metals_best_conservative_no_norm_all_aat.sh	me	Jun 2 me	4 KB	
run_best_conservative_no_norm_all_aat.sh	me	Jun 2 me	5 KB	

Dr. Joanne Hughes
Associate Professor
Physics Department
Seattle University

From: Joanne Hughes Clark <jhughes@seattleu.edu>

Sent: Monday, June 15, 2026 9:42 AM

To: Joseph McEwen <jmcewen@seattleu.edu>; Payton Killcoyne <pkillcoyne@seattleu.edu>; Thea Campbell <tcampbell@seattleu.edu>; Laith Rabaa <lrabaa1@seattleu.edu>; Michael Williams <mwilliams8@seattleu.edu>; Keleb Mwaala <kmwaala@seattleu.edu>

Cc: Kerry Cheon <kcheon@seattleu.edu>

Subject: Re: Summer research update

Hi Folks,

Make sure you have a Google account (costs nothing) to see my spectroscopy workflow. All the files should be here:

<https://drive.google.com/drive/folders/1puntGctkDdHnERWG02Lgjt8Wd9fvtRqK?usp=sharing>

Download all the files and store them somewhere before you start changing them.

These are Jupyter Notebooks that run on my home PC, so you have to create directories and name them sensibly, replacing my directories with yours in the Notebooks.

Cheers,
Joanne

Dr. Joanne Hughes
Associate Professor
Physics Department
Seattle University

From: Joanne Hughes Clark <jhughes@seattleu.edu>

Sent: Thursday, June 11, 2026 12:07 PM

To: Joseph McEwen <jmcewen@seattleu.edu>; Payton Killcoyne <pkillcoyne@seattleu.edu>; Thea Campbell <tcampbell@seattleu.edu>; Laith Rabaa <lrabaa1@seattleu.edu>; Michael Williams <mwilliams8@seattleu.edu>; Keleb Mwaala <kmwaala@seattleu.edu>

Cc: Chris Varney <varneych@seattleu.edu>; Kerry Cheon <kcheon@seattleu.edu>

Subject: Re: Summer research update

Okay-consensus is we will meet next Wednesday. I'll be in about 10:05, so we can meet anytime after 10:30.

Best,
Joanne

Dr. Joanne Hughes
Associate Professor
Physics Department
Seattle University

From: Joanne Hughes Clark <jhughes@seattleu.edu>

Sent: Thursday, June 11, 2026 8:30 AM

To: Joseph McEwen <jmcewen@seattleu.edu>; Payton Killcoyne <pkillcoyne@seattleu.edu>; Thea Campbell <tcampbell@seattleu.edu>; Laith Rabaa <lrabaa1@seattleu.edu>; Michael Williams <mwilliams8@seattleu.edu>; Keleb Mwaala <kmwaala@seattleu.edu>

Cc: Chris Varney <varneych@seattleu.edu>

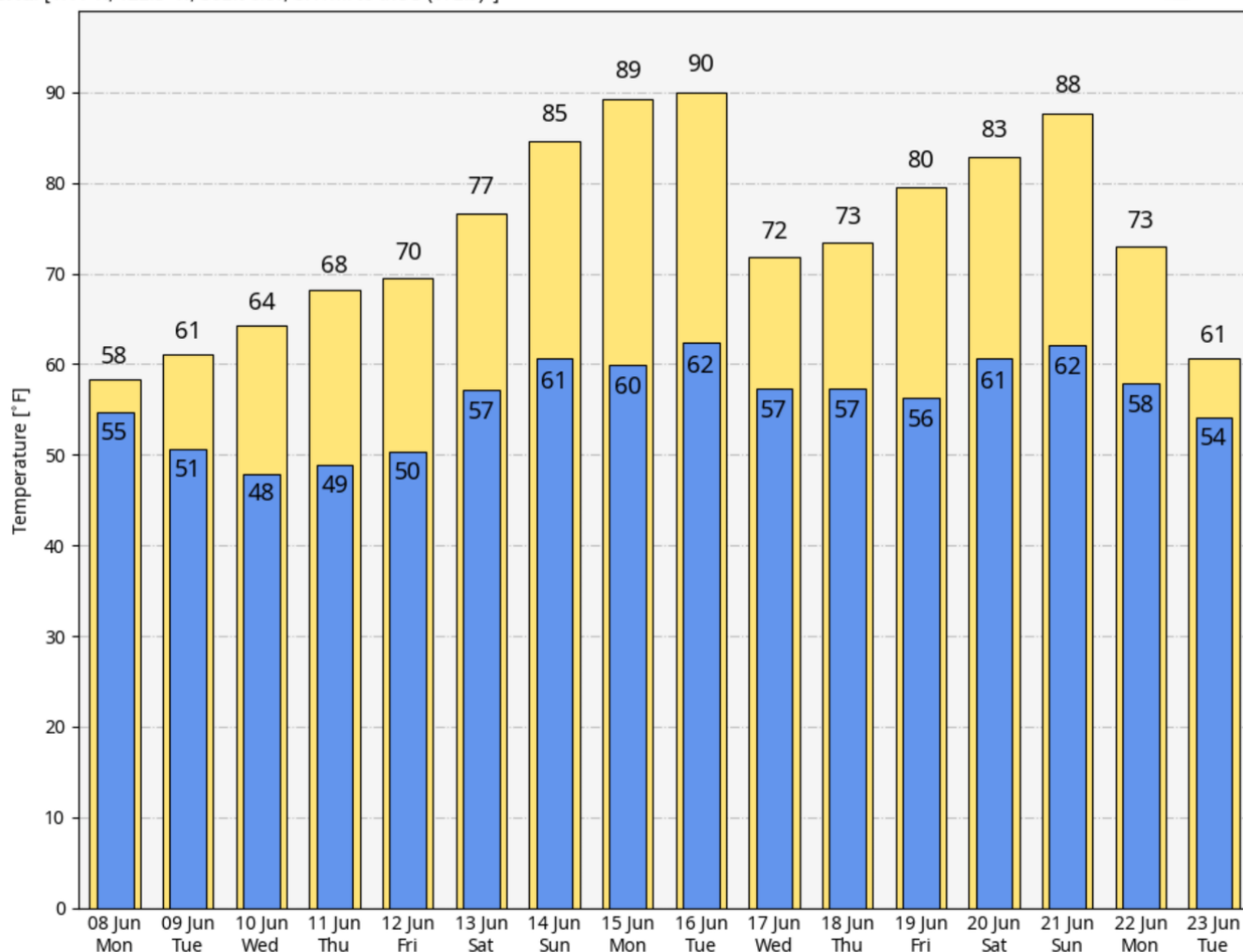
Subject: Summer research update

Hi Folks,

I'm assuming our workroom is Bann 148 (same as last year). I want to make sure we have what we need and maybe have one of the huge monitor-TVs in there so we can share work. I will also update the Canvas "course" and add any Zoom meetings.

Are we still planning to meet on Monday? The reason I ask is that Monday and Tuesday are supposed to reach 90F and Bannan has no AC (my office is worse than the lab) because of the windows. I'm suggesting we start Wednesday in person, and have everybody make sure their computers are running the right environments before that. I propose we Zoom at 10:00 on Monday to start and meet at 10:30 on Wednesday in person. This work is mostly computer-based, so it can be done remotely, but we are going to need to meet as a group a few times per week to keep everybody on task. We do have group computers in the workspace, so it isn't urgent if your personal laptop or desktop is uncooperative.

ECMWF 0.1° Init 12z 8 Jun 2026 • Daily High/Low Temperature (°F)
Seattle Tacoma Int'l Airport • KSEA [47.449°N, 122.309°W, 433ft elev]
Grid: [47.4°N, 122.3°W, 302ft elev, 3.41mi to the S (172.9°)]



For the new students, make sure you have something like Ubuntu 20.04.3 LTS (GNU/Linux 4.4.0-19041-Microsoft x86_64) on your PC, and this is just what I did a few years ago, but I'm experienced with it. We need to get everybody's systems set up to work the same way before tackling the research.

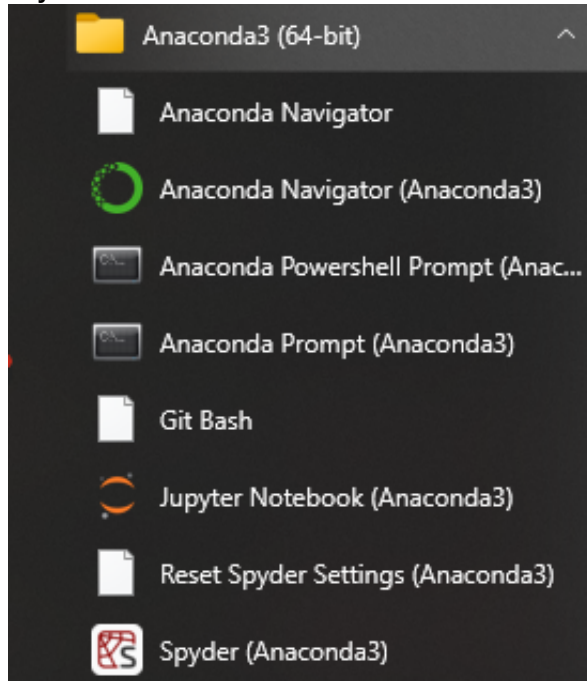
The Anaconda Distribution comes pre-packaged with Python, Jupyter Notebook, and hundreds of popular data science libraries.

1. **Download Anaconda:** Visit the official website and download the free installer from the

Anaconda Downloads Page.

2. **Run the Installer:** Double-click the downloaded setup file.
3. **Follow the Prompts:** Click **Next**, accept the license terms, and select **Just Me** or **All Users**. Stick to the default installation path.
4. **Launch Jupyter:** Open your Start menu, look for **Anaconda Navigator**, and click **Launch** under the Jupyter Notebook icon. Alternatively, open the **Anaconda Prompt**, type `jupyter notebook`, and hit Enter.

My menu looks like this:



This is the minimum needed to start the spectroscopy project. I normally pop up 2 Anaconda Powershells and type "bash" to get into the right shell.

I normally then start the Anaconda Navigator. In my system, the command to run Python on a script is: "python3 run.py". I prefer to use the Jupyter Notebook most of the time, and it is where I create the main data analysis scripts. For outside code, I use Google Colab, so make yourself an account. I do not use Claude, which Joe does, but this is up to the individual. I have a ChatGPT subscription to help me code in Python, since I was trained in FORTRAN. You will need FORTRAN and C compilers, which I have included in these instructions.

I copied the instructions from Google about how to get started:

To configure your PC for Python and Jupyter Notebooks, you can choose between two primary paths: **using the all-in-one Anaconda distribution (easiest for beginners) or manually installing Python, virtual environments, and Jupyter via PIP (best for lightweight and custom setups)**. [\[1, 2, 3\]](#)

Method 1: The All-in-One Anaconda Method (Recommended for Beginners) [\[4, 5, 6\]](#)

The Anaconda Distribution comes pre-packaged with Python, Jupyter Notebook, and hundreds of popular data science libraries.

1. **Download Anaconda:** Visit the official website and download the free installer from the Anaconda Downloads Page .
2. **Run the Installer:** Double-click the downloaded setup file.
3. **Follow the Prompts:** Click **Next**, accept the license terms, and select **Just Me** or **All Users**. Stick to the default installation path.
4. **Launch Jupyter:** Open your Start menu, look for **Anaconda Navigator**, and click **Launch** under the Jupyter Notebook icon. Alternatively, open the **Anaconda Prompt**, type `conda activate base`, and hit Enter. [[1](#), [7](#), [8](#), [9](#), [10](#)]

Method 2: The Lightweight Manual Method (For Experienced Users)

If you prefer to keep your machine free from large distributions, you can configure Python and Jupyter directly using your system's Command Prompt or PowerShell . [[2](#), [11](#), [12](#)]

Step 1: Install Python

1. Download the latest installer from the official Python Downloads Page .
2. Run the executable file.
3. **Crucial:** Check the box that says "**Add python.exe to PATH**" before clicking **Install Now**.
4. Verify the setup by opening your Command Prompt () and running: [[11](#), [13](#), [14](#)]

Step 2: Create a Virtual Environment (Best Practice)

Isolated environments keep your project libraries from interfering with each other.

1. Open your Command Prompt, make a project folder, and navigate inside it:
2. Generate the environment:
3. Activate it:
 - **Windows:**
 - **Mac/Linux:** [[2](#), [19](#)]

Step 3: Install Jupyter Notebook via PIP

1. While your environment is active, upgrade :
2. Install the classic Jupyter interface using the Project Jupyter package manager:
3. Boot up the application: [[20](#), [21](#), [22](#), [23](#)]

Method 3: Integrated Visual Studio Code Setup

If you prefer working inside a dedicated code editor, you can run Jupyter Notebooks directly within Microsoft's editor.

1. Download and install the software from the VS Code Official Site .
2. Click the **Extensions** icon on the left sidebar (or press `Ctrl+Shift+X`).
3. Search for and install the **Python** extension by Microsoft.
4. Search for and install the **Jupyter** extension by Microsoft.
5. Create a new file, save it with the extension (e.g., `.py`), and select your Python environment as the top-right active **Kernel**. [[2](#), [12](#), [14](#), [24](#), [26](#)]

Would you like help deciding **which setup option fits your specific PC hardware**, or do you need assistance **installing specific libraries like Pandas or NumPy** once it's up?

AI responses may include mistakes.

- [1] <https://www.youtube.com/watch?v=PxMN8xT9NbA>
- [2] <https://www.youtube.com/watch?v=mTIY6MC58J8>
- [3] <https://www.youtube.com/watch?v=sNu2P69cojk>
- [4] <https://www.jaroeeducation.com/blog/how-to-install-jupyter-notebook>
- [5] <https://www.trymito.io/blog/how-to-install-jupyter-notebook>
- [6] https://people.minesparis.psl.eu/fabien.moutarde/ES_MachineLearning/Practical_clustering/practical_clustering.html
- [7] <https://www.codecademy.com/article/install-jupyter-notebook-on-mac-and-windows>
- [8] <https://medium.com/@neuralnets/beginners-quick-guide-for-handling-issues-launching-jupyter-notebook-for-python-using-anaconda-8be3d57a209b>
- [9] <https://opentechschoool.github.io/python-data-intro/core/recap.html>
- [10] <https://community.mis.temple.edu/mis3536sec001spring2025/files/2025/02/Quick-Guide-to-Jupyter-Notebook-Python-and-Anaconda.pdf>
- [11] https://www.youtube.com/watch?v=HLD-LI_-IT4
- [12] <https://learn.microsoft.com/en-us/windows/dev-environment/python>
- [13] <https://www.youtube.com/watch?v=0z493LWLWQw>
- [14] <https://www.youtube.com/watch?v=HeghQMzzXH8>
- [15] <https://digipedia.tudelft.nl/tutorial/jupyter-notebook-installation-and-overview/>
- [16] https://patrickclarke.ca/blog/python_virtual_environments.html
- [17] <https://www.digitalocean.com/community/tutorials/install-python-windows-10>
- [18] <https://apxml.com/courses/introduction-to-computer-vision/chapter-1-getting-started-computer-vision/setting-up-development-environment>
- [19] <https://www.pyquantnews.com/free-python-resources/install-python-and-set-up-development-on-windows-macos-linux>
- [20] <https://docs.jupyter.org/en/latest/install/notebook-classic.html>
- [21] <https://jupyter.org/install>
- [22] <https://medium.com/mr-plan-publication/create-python-virtual-environment-e6286e4f3009>
- [23] https://radcamp.github.io/AF-Biota/Jupyter_Notebook_Setup.html
- [24] <https://code.visualstudio.com/docs/datascience/jupyter-notebooks>
- [25] <https://fsc.stevens.edu/installing-python-on-windows-11-workstations/>
- [26] <https://www.youtube.com/watch?v=nO2T2IRFI50>

If you have a Mac, this is how you should start a new environment:

You can set up Jupyter Notebook on a Mac using two primary methods: **using a Python virtual environment** (highly recommended for a clean setup) or **using Anaconda** (best for data science beginners). [1]

Method 1: Clean Setup using Pip & Virtual Environments (Recommended)

This approach prevents conflicts with your Mac's built-in system files by isolating your Jupyter environment. [2, 3, 4, 5]

1. Check or Install Python 3

Open your **Terminal** app (Press `⌘`, type "Terminal", and hit Enter) and verify you have Python

3 installed: [\[3, 6\]](#)

If you see an error or an outdated version, download and run the macOS installer from the Official Python Downloads Page. [\[6, 7\]](#)

2. Create a Project Folder and Virtual Environment

Navigate to your desired directory and build a self-contained workspace: [\[2, 3\]](#)

3. Activate the Environment

Turn on your virtual environment. Your terminal prompt will display to show it is active: [\[2, 3, 8, 9\]](#)

4. Install Jupyter Notebook

Upgrade your package installer and download Jupyter directly into your active workspace: [\[8, 10\]](#)

5. Launch the Notebook

Run this command to fire up the interface: [\[8, 11, 12\]](#)

A new tab will automatically open in your web browser at displaying your current directory. [\[6, 13, 14, 15, 16\]](#)

Method 2: All-in-One Setup using Anaconda (Easiest for Data Science) [\[10, 17\]](#)

If you prefer a visual user interface that comes pre-packaged with libraries like NumPy, Pandas, and Matplotlib, use Anaconda.

1. Go to the Anaconda Downloads Page .
2. Choose the graphical installer matching your Mac's processor chip:
 - **Apple Silicon** (M1, M2, M3, etc.)
 - **Intel** (Unsure? Click the Apple logo in the top-left corner of your screen and select About This Mac to check). [\[10, 18, 22\]](#)
3. Double-click the downloaded file and follow the step-by-step installation instructions.
4. Once completed, open **Anaconda Navigator** from your Applications folder.
5. Find the **Jupyter Notebook** icon and click **Launch**. [\[18\]](#)

Creating Your First Notebook

Once Jupyter opens in your web browser:

1. Look to the top right corner, click the **New** dropdown menu, and select **Python 3 (ipykernel)**.
2. In the empty gray cell that appears, type your code, for example:
3. Press **Shift + Enter** on your keyboard to run the code cell and view your output instantly.
4. To shut down your workspace when finished, go back to your Terminal window and press **Ctrl + C** twice. [\[3, 6, 7, 18, 24\]](#)

- You need to install specific **data packages** (like NumPy, Pandas, Astropy, or PyTorch)

I can guide you through the exact setup commands for your workflow.

AI responses may include mistakes.

- [1] https://www.reddit.com/r/pythontips/comments/1cx7ky1/jupyter_notebooks_on_mac/
- [2] https://www.reddit.com/r/learnpython/comments/1sar4ha/how_to_properly_install_pythonjupyter_notebook_on/
- [3] <https://www.youtube.com/watch?v=pkjtbnsX7Yw>
- [4] <https://saturncloud.io/blog/running-jupyter-notebook-in-a-virtual-environment-installed-scikitlearn-module-not-available/>
- [5] <https://linuxcapable.com/how-to-install-jupyter-notebook-on-ubuntu-linux/>
- [6] <https://www.youtube.com/watch?v=2X9c1-VNibc>
- [7] <https://www.youtube.com/watch?v=z7zOkRublrU>
- [8] <https://www.runcell.dev/blog/install-jupyter-notebook-mac>
- [9] <https://medium.com/@anchitgupt/the-complete-guide-to-running-apache-spark-with-jupyter-notebook-on-your-macbook-e8deee534eb8>
- [10] <https://www.youtube.com/watch?v=PUNUO4Eo6Ac>
- [11] <https://jupyter.org/install>
- [12] <https://setapp.com/how-to/how-to-install-jupyter-notebook-on-mac>
- [13] <https://jupyter-notebook-beginner-guide.readthedocs.io/en/latest/execute.html>
- [14] <https://afox.land/post/jupyter-anaconda/>
- [15] <https://pyar.ucsc.edu/python-tutorial-session/>
- [16] <https://www.udacity.com/blog/jupyter-notebook-tutorial-a-beginners-guide-for-the-data-science-tool/>
- [17] <https://python.plainenglish.io/how-to-install-python-for-data-analytics-step-by-step-guide-to-anaconda-miniconda-pip-23065bd5a376>
- [18] https://www.youtube.com/watch?v=I_by03efxyw
- [19] <https://daily.dev/blog/jupyter-for-beginners/>
- [20] <https://melaniewalsh.github.io/Intro-Cultural-Analytics/02-Python/01-Install-Python.html>
- [21] <https://www.debugpoint.com/jupyter-notebook-tutorial/>
- [22] <https://rcdsdocs.it.northwestern.edu/tutorials/python/python-installation.html>
- [23] <https://www.codecademy.com/article/install-jupyter-notebook-on-mac-and-windows>
- [24] <https://www.youtube.com/watch?v=tblg7MzOqlk>

Configuring a PC for Python and Fortran involves installing a Fortran compiler (such as), installing Python, and setting up Visual Studio Code as your IDE. To bridge the two, you will use to compile Fortran code into high-speed Python modules. [1, 2, 3, 4, 5, 6]

Follow these concrete steps to get your PC ready:

1. Install Fortran and C Compilers

Fortran needs a compiler to translate your code into machine-readable files.

- **Windows:** Use MSYS2. Once installed, open the MSYS2 terminal and run: to install the GNU Fortran compiler (). Make sure to add the compiler's folder to your Windows System variable.

- **macOS:** Open terminal and run using Homebrew.
- **Linux (Ubuntu):** Open terminal and run . [2, 4, 7, 8, 9]

2. Install Python

Python will act as your high-level environment for organizing data and executing your programs.

- Go to the official Python Downloads page and grab the latest version.
- **Crucial for Windows:** During the installer setup, ensure you check the box that says "Add python.exe to PATH" before clicking "Install Now". [5, 12, 13]

3. Set Up Visual Studio Code (VS Code)

VS Code will serve as your primary code editor.

- Download and install the editor from Visual Studio Code .
- Open the Extensions tab (Sidebar) and install these essential packages:
 - **Python** (by Microsoft)
 - **Modern Fortran** (by Fortran-Lang)
 - **C/C++** (by Microsoft, required for debugging Fortran tools)

4. Interface Fortran and Python

To run heavy numerical calculations using Fortran while driving the process in Python, use the tool, which comes bundled with Python.

- Create a Fortran file (e.g.,).
- Open your terminal and use the following command to automatically generate a Python-importable library:
- You can now open a Python script and type to utilize your high-speed Fortran subroutines natively. [3, 5]

Further Exploration

- Explore advanced configuration and environment variable setup via the Matthias Noback Blog.
- Reference the official Fortran OS Setup Guide for comprehensive installation procedures across various operating systems.
- Join community discussions regarding complex packaging and static linking on the Fortran Discourse Forum. [14, 15, 16]

Could you tell me what kind of project you are working on? Knowing if you are doing **data analysis**, **numerical simulations**, or **machine learning** will help me guide you to the right packages (like NumPy or SciPy) to pair with your Fortran code.

AI responses may include mistakes.

[1] <https://www.youtube.com/watch?v=s1pC2Ini1rl>

[2] <https://www.youtube.com/watch?v=5CPhf-GqWb0>

- [3] <http://mitran-lab.amath.unc.edu/courses/MATH662/examples/f2py/f2py-1.pdf>
- [4] https://www.reddit.com/r/fortran/comments/yzmud9/how_to_install_fortran_on_windows/
- [5] <https://www.youtube.com/watch?v=w57h4nqy7il>
- [6] <https://www.youtube.com/watch?v=RrsoM6wVEWE>
- [7] https://fortran-lang.org/learn/quickstart/hello_world/
- [8] <https://www.youtube.com/watch?v=QdfVAO1DFus>
- [9] <https://fortranwiki.org/fortran/show/GFortran>
- [10] <https://www.youtube.com/watch?v=sWQowh6-PCg>
- [11] <https://www.springboard.com/blog/software-engineering/how-to-become-a-python-developer/>
- [12] <https://www.youtube.com/watch?v=NfTrFhfqmFY>
- [13] <https://www.youtube.com/watch?v=IQ2LVbFSenQ>
- [14] <https://matthiasnoback.nl/2025/05/running-a-simple-fortran-program/>
- [15] <https://fortran-lang.discourse.group/t/interfacing-fortran-code-from-python/1280>
- [16] https://fortran-lang.org/learn/os_setup/

Hope your exams went well.

Joanne

Dr. Joanne Hughes
Associate Professor
Physics Department
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